DRUGS · ANTIBIOTICS: OTHER CLASSES

Co-trimoxazole

Co-trimoxazole combines trimethoprim and sulfamethoxazole, two drugs that block successive steps in bacterial folate synthesis. Its ENT roles are specific: maintenance of remission in localised granulomatosis with polyangiitis (and reducing the nasal Staphylococcus aureus carriage that triggers relapse), Pneumocystis prophylaxis in immunosuppressed patients, and oral treatment of some MRSA and resistant infections. It is not a first-line drug for routine ENT infection because of its toxicity.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

CO-TRIMOXAZOLE

	SUMMARY
Class	Sulfonamide with a dihydrofolate reductase inhibitor
How it works	Sulfamethoxazole blocks dihydropteroate synthase; trimethoprim blocks dihydrofolate reductase; synergistic and bactericidal
Spectrum	S. aureus including many MRSA, H. influenzae, Moraxella, Enterobacterales, Stenotrophomonas, Pneumocystis jirovecii, Nocardia; streptococci variable; not Pseudomonas, not anaerobes
ENT uses	Localised (limited) granulomatosis with polyangiitis, adjunct to maintenance; Pneumocystis prophylaxis during high-dose steroids, cyclophosphamide or rituximab; MRSA skin and soft tissue infection; Nocardia
Adult dose	960 mg twice daily for treatment; 960 mg twice daily in GPA maintenance (some use 480 mg twice daily); Pneumocystis prophylaxis 480 mg once daily or 960 mg three times a week
Child dose	24 mg/kg twice daily for treatment; prophylaxis by body surface area on specialist advice

	SUMMARY
Avoid	Folate deficiency, blood disorders, severe liver or kidney disease, porphyria, first trimester pregnancy, G6PD deficiency (haemolysis)

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

Bacteria must make their own folate; blocking two steps in the pathway gives synergy. Human dihydrofolate reductase is far less sensitive to trimethoprim, but toxicity rises in folate-deficient patients and with methotrexate.

2.2 Handling

Well absorbed; renally excreted, so the dose is reduced and potassium monitored in renal impairment.

3 CHAPTER THREE Using it in ENT

Indications and doses.

3.1 Granulomatosis with polyangiitis

In limited sinonasal GPA, co-trimoxazole may help maintain remission and reduce relapses, possibly by eradicating nasal S. aureus; it does not replace immunosuppression in generalised disease. Patients on rituximab or cyclophosphamide receive it as Pneumocystis prophylaxis.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- Rash, including **Stevens-Johnson syndrome** and toxic epidermal necrolysis (stop at the first rash).
- **Bone marrow suppression** (leucopenia, thrombocytopenia, megaloblastic anaemia), especially in the elderly and folate deficient.
- **Hyperkalaemia** (trimethoprim acts like amiloride), raised creatinine without a true fall in renal function.
- Hepatitis, aseptic meningitis, haemolysis in G6PD deficiency.

4.2 Interactions

Methotrexate (severe, potentially fatal marrow toxicity), **warfarin** (raised INR), ACE inhibitors, spironolactone and other potassium-raising drugs, phenytoin, ciclosporin, azathioprine.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

> **WHAT THE EXAMINER IS LOOKING FOR**
>
> - Its role in **GPA**: adjunct in limited disease and Pneumocystis prophylaxis.
> - Haematological, skin and potassium monitoring.
> - The **methotrexate** and **warfarin** interactions.

5.2 Common pitfalls

COMMON PITFALLS

- Using co-trimoxazole for routine sinusitis or tonsillitis.
- Combining it with methotrexate.
- Ignoring hyperkalaemia in a patient on an ACE inhibitor.

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